

[DC] Virtual Reality Nature Environment Designs for Mental Health

Rachel Masters*

Colorado State University

ABSTRACT

Shinrin-yoku, or forest bathing, is a therapeutic nature immersion practice shown to reduce stress and restore mental resources. In a world where the negative effects of sustained stress are an increasingly prevalent issue, connection to nature is important. However, many populations who encounter the most stress are largely isolated from nature, like people working long hours in cities, nursing home residents, or hospital patients. For these populations, virtual reality (VR) nature immersion is a promising supplement for when nature is not accessible. In order to design optimal VR nature environments (VNEs), we must first understand why and how VNEs can be restorative. This is a current area of research, and one critical gap that the proposed research seeks to understand is how the different components of VNEs and their interactions can be optimally designed for short and long term stress reduction and attention restoration. The proposed research does a deeper exploration into the designs of plants and water for optimally restorative VR green and blue spaces.

Index Terms: Human-centered computing—Empirical Studies in HCI—; —Human-centered computing—User studies—; Human-centered computing—Virtual Reality—

1 INTRODUCTION

The damaging effects of stress on physical and mental wellbeing is an under addressed yet prevalent problem. An effective solution for stress reduction is forest bathing, but nature access is limited for many people. Virtual reality (VR) is an accessible platform that can create immersive, lifelike experiences. My research will identify essential environmental components for effective virtual nature environments (VNEs) that mitigate negative effects of stress both short and long-term. Stress affects the bodily systems that control how people behave, think, and feel, damaging one's overall quality of life. [13] Additionally, in high stress environments, people often experience cognitive overload, which is a depletion of mental resources due to being mentally overwhelmed. [13] Forest bathing (FB), a therapeutic technique involving immersion in nature, is an effective solution for both reducing stress and restoring mental resources. [5] The problem with FB is that it is not always an accessible solution. For example, urban life is associated with both general and chronic stress, [11] yet people in cities often have less access to nature despite having the greatest need. Urban dwellers need a uniquely immersive, accessible platform that can provide benefits similar to nature, and VR is a promising option. As portable VR technologies like the Oculus Quest 2 become more accessible and affordable, virtual stress relief tools have the potential to be reliable supplements. Since VR cannot fully replicate reality, it is critical to understand the essential components for creating effective VNEs. In the fall of 2021, I conducted initial research on the restorative impact of biomass, which is living green plant life. Results indicated that biomass is a potentially important component in restorative

virtual environments, but more research is needed to elucidate the role of biomass in promoting restoration and more comprehensively investigate other characteristics of the environment design. [8] I propose investigating how VR implementations of key natural features can affect stress reduction and restoration in both the short and long-term.

2 RESEARCH PLAN

This research plan has three phases. Phase one focuses on identifying the importance of biomass implementations for the restorative effect. Phase two investigates the importance of water implementations for the restorative effect. Phase three investigates longer term effects of VNEs in office and educational spaces. For phases one and two, psychological measures follow those from Masters et. al. [8], including the Positive and Negative Affect Schedule and Perceived Restorativeness Scale, and adding the Perceived Stress Scale [2]. Additionally, electrocardiogram (ECG) and blood pressure are measured, and the Trail Making Test [10] is used to measure attention. Best measures for the phase three longevity studies will be determined from prior work. Before phase one, a review of VR FB literature from the past five years will be conducted to understand the current state of VR FB research and inform the rest of the proposed research.

2.1 Phase 1

Phase one is a two-part study on biomass implementations. It is largely unknown how plant biodiversity affects the restorative potential of VNEs. In real nature environments, biodiversity has a variety of health benefits; primarily related to diverse microbiomes and their effects on physical health [7]. Since microbiomes cannot be modelled in VR, a deeper understanding of how visual plant biodiversity affects mental health in VR is essential for creating VNEs with the visual requirements to maintain passive attention — a key element of perceived restorativeness [4]. Additionally, the PRS, a common measure of attention restoration in natural environments, measures how well an environment is an escape from everyday life [4]. It is possible that only modeling green and blue spaces in VR may be eliminating other nature options that are much more difficult to find in real life, and that creating natural environments that people do not see often in real life may also have restorative effects. We aim to answer the question: is biomass required to be green for the restorative effect? Color theorists suggest that green, blue, and purple have relaxing qualities [3].

Study 1: The first study examines the question: to what degree does biomass diversity influence the efficacy of a VNE? There are two levels, one with homogenous biomass and one with diverse biomass. Participants are randomly assigned to each environment, experiencing the environment for 10 minutes after undergoing the Trier Social Stress Test. The hypothesis is that environments with greater biomass diversity are more restorative than environments with homogenous biomass because more different visually interesting environments should be better at capturing passive attention [5].

Study 2: The second study examines the question: is biomass required to be real and green for the restorative effect? There will be an environment with realistic, green biomass, and one with purple

*e-mail: ramast1@colostate.edu

biomass. The experiment design flow is the same as the first study. Research on ‘nearby nature,’ or nature in everyday life, suggests that green plants specifically provide restorative benefits, so the hypothesis is that the green environment is more restorative [5].

2.2 Phase 2

Blue spaces, or spaces with water access, have shown potential to be restorative like green spaces, both with respect to real nature [1] and in VR [12], but optimal design of these blue spaces in VR has yet to be investigated. Understanding which blue spaces are most effective for restoration and how water can be best designed in VR is essential for the design of optimal VNEs.

Study 1: The first step to understanding blue spaces is to understand how the interaction of biomass and water on restorative affect as well as the unique effects of blue spaces compared to green biomass. The research question we aim to answer is: does water influence the efficacy of a VNE? There will be four environments with the same structural layout, with and without water combined with biomass or no biomass. We will conduct a between-subjects experiment on these environments, with a between-subjects control condition added, and the experimental design flow and metrics will be the same as Section 2.1. Based on results from prior literature showing preference for blue spaces [12], our hypothesis is that the environments with water are more restorative than the environments without water, and that the biomass environment with water is the most restorative, due to the restorative nature of blue and green spaces.

Study 2: After understanding the effects of the presence of water, the next step to understanding optimal blue space design is to understand the perception of water movement and what types of water movement are most restorative. There are four environments for this experiment, implemented using the most effective environment from the first water study as the foundation: one with a still pond, one with a rippling pond, one with a still river, and one with a flowing river. The experiment design will be the same as the last water experiment. Our hypothesis is that the moving water is more restorative than still water, and a flowing river is most restorative because the biophilia hypothesis suggests that humans resonate with living, moving nature [6].

2.3 Phase 3

Phase three is a two-part study on longer term effects of VNEs, answering the question: are VNEs effective for decreasing stress over time? The long-term effects of VNEs are little researched, yet VNEs are intended to have long-term, everyday use cases in workplaces and homes that need preliminary observation and fine tuning for effectiveness before widespread public use. Two longitudinal studies are to be conducted, one in a workspace, and one in a university educational space on student stress. These longevity studies are conducted by deploying the best natural environments from previous experiments on the Meta Quest 3 and integrating them into people’s work and study habits for two months, observing long term responses using the psychological questionnaires. These studies provide deeper insight into the efficacy of the VNEs because the stress induced by lab stressor tests do not fully model the general and chronic stress experienced in everyday life, which is observed in these studies.

3 CONTRIBUTIONS

This research will advance the field of Human Computer Interaction (HCI) by providing deeper insights on how VR can be designed to affect Perception in positive ways. Specifically, this research will further knowledge of how VNEs can be designed to help people improve mental health via stress reduction and cognitive resource restoration. These VNEs can be used to mitigate chronic stress in workplaces, educational spaces, and homes. Additionally, this

research will provide an assessment of the current state of VR FB literature and will advance knowledge of VR FB, which can be used in simulations to model city park natural areas before creating them. This research will inform the design of optimal VNEs, after which new interdisciplinary applications of these VNEs can be explored. These results combined with future research efforts can make VR an accessible tool to improve users’ mental health.

4 CONCLUSION

This research will provide deeper insights into the intricacies of VNE component design for the optimal promotion of mental health. I have completed the literature review mentioned in Section 2, have created an experiment design for the first experiment in Section 2.1, and am designing the environments to test in 2024. Multiple limitations pose potential challenges to this work, including cybersickness and limited understanding of how personal preference affects experience. Open response questions to better understand these topics have been added to the proposed experiments. Additionally, two main areas of improvement relate to the success of the proposed research. How can environments be designed to maximize immersion and presence? How can multiple senses be incorporated to derive the most benefit from VNEs? These are key questions in current literature, and future work aims to address these topics in more detail [9].

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